

Eskandar (Alex) Seif

Software Architect & Full-Stack Developer

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PROFESSIONAL SUMMARY

Pragmatic Software Architect and Full-Stack Developer with over 20 years of experience designing, scaling, and maintaining high-availability web applications and transactional infrastructure. Proven track record taking mission-critical systems from initial architecture to production, including national-scale public sector platforms serving millions of citizens, region-first online payment gateways, and decoupled spatial GIS engines. Deep specialist in the PHP ecosystem (Symfony, Drupal Enterprise, WordPress Core/FSE/ACF, Moodle), JavaScript/TypeScript (React, Node.js), Python (FastAPI), and relational database architecture (MySQL 3NF/Star Schema replication, PostgreSQL/PostGIS). Focuses on pragmatic software architecture, zero-downtime database migrations, deterministic DevOps automation, and eliminating technical debt.

TECHNICAL STACK & ARCHITECTURE COMPETENCIES

Architecture & System Design:	Distributed Systems, Decoupled Monoliths, High-Concurrency Routing, Domain-Driven Design (DDD), Transactional State Machines, REST & OpenAPI Contract Synchronization.
PHP Ecosystem:	Symfony Framework, Drupal Enterprise (Core, 8/9/10), WordPress (VIP, Multisite, Full Site Editing, ACF Pro, Gutenberg API), Moodle LMS, Custom MVC Frameworks, PHP 8.x / Legacy 7.x.
Full-Stack & Frontend:	TypeScript, React (React 19, Hooks), JavaScript (ES6+), Node.js, TailwindCSS, Vite, HTML5, CSS3 / SCSS.
Python & Geospatial:	Python 3.12, FastAPI, SQLAdmin, PostgreSQL 15+, PostGIS (Spatial Indexing, SRID 4326, GeoJSON).
Database & Data Layer Optimization:	MySQL / MariaDB (Master-Slave Replication, Query Plan Optimization, 3NF Transactional Normalization, Star Schema Analytical Views, Indexed Views), Redis.
DevOps, Cloud & Infrastructure:	Docker / Docker Compose, Linux Administration (CentOS, Oracle Linux, Debian), Nginx (Reverse Proxy, Media Streaming), AWS (EC2, ELB, S3, CloudFront), CI/CD (GitHub Actions, Jenkins), Air-Gapped Offline Deployment, Bash Automation.
Identity & Security Compliance:	SAML 2.0 Federation, Keycloak SSO, PCI DSS Compliant Architecture, SAMA Financial Compliance, Role-Based Access Control (RBAC).

FEATURED ARCHITECTURAL ENGAGEMENTS (DEEP DIVES)

barincairo.com (2026) (Software Architect & Full-Stack Developer) Architected a decoupled Web GIS platform (React 19 SPA, Python 3.12 FastAPI, PostGIS) featuring sub-second spatial viewport queries, automated multi-tier web scraping pipelines, and Docker containerized deployment.

Greek Community of Alexandria Portal Modernization (2026) (Software Architect & Full-Stack Developer) Programmatically decoupled and modernized a legacy CMS monolith into a lightweight, zero-dependency Full Site Editing architecture via script-first AST parsing workflows with 1:1 visual fidelity.

North South Consultants Exchange (NSCE) (2025) (Software Architect & Full-Stack Developer) Architected an institutional project impact and donor transparency platform used by international development funders (UN, GIZ, EU) to evaluate multi-country programs across the MENA region.

Region-First Payment Gateway (Walleety) (2009) (Co-Founder & Software Architect) Architected and deployed Egypt's first independent online payment gateway from scratch. Engineered asynchronous CLI queue daemons and defensive transactional state machines, scaling the platform to process 33% of domestic airline ticketing transactions.

PROFESSIONAL EXPERIENCE

SOFTWARE ARCHITECT & FULL-STACK DEVELOPER

iSoft (IdealSoft Emirates) | Retainer & Contract2020-2024

Contracted by iSoft to deliver ministry-level digital infrastructure, high-concurrency transactional routing engines, and API architectures for federal and municipal public sector clients across Egypt and the GCC region.

- **National Vehicle Inspection Scheduling Infrastructure (SASO - Saudi Arabia):** Architected the nationwide vehicle inspection booking system managing multi-lane facilities. Engineered a distributed transactional state-locking engine with a deterministic 5-minute buffer that eliminated race conditions across competing web, mobile, and backend channels for identical 15-minute appointment slots. Remodeled database schemas into 3NF for transactional safety and Star Schema for real-time reporting. Decoupled Drupal core to expose versioned REST APIs to a high-performance React client.
- **Air-Gapped Intranet Certification Platform (Egyptian Ministry of Interior - Traffic Police Dept.):** Architected and deployed a highly locked-down enterprise training portal on a restricted state intranet for mandatory licensing courses nationwide. Engineered a deterministic zero-network deployment pipeline using a custom-patched Oracle Linux OS repository on physical USB storage with fault-tolerant Bash migration scripts and offline cryptographic fee verification.
- **National Vehicle Inspection Digital Ecosystem (Aman Portal):** Designed the digital booking and dispatch engine for nationwide vehicle inspections, redirecting citizen volume from congested municipal police departments to commercial hubs and mobile inspection vans via high-availability backend API integration.
- **State-Licensed Cooperative Insurance Platform (Salama Portal - Saudi Arabia):** Engineered the core registration workflows, pre-ingestion validation mechanics, and secure transaction-handling layers for compliant digital automotive insurance under SAMA regulatory standards.

SOFTWARE ARCHITECT & FULL-STACK DEVELOPER (CONTRACT)

Meem Development (SEF Academy)January 2024 - December 2024

- **SAML 2.0 & Keycloak SSO Federation:** Re-engineered metadata schemas and SAML attribute assertions between Keycloak and Moodle, resolving authentication failures and stabilizing corporate Single Sign-On.
- **Secure Database Topology:** Configured role-isolated SSH tunneling protocols for distributed engineering teams, enabling secure concurrent database operations on Test and UAT instances without public port exposure.
- **Transactional Email & Integrations:** Configured authenticated SMTP relays for deterministic course certificate dispatch and validated enterprise Microsoft Teams integration.

CO-FOUNDER & SOFTWARE ARCHITECT

Walleyt (Fintech Platform)December 2009 - July 2015

Architected, launched, and scaled an early online payment gateway in Egypt, processing 33% of all domestic air travel ticketing volume.

- **Core Engine Delivery:** Solely engineered the core payment gateway using PHP and Symfony, resolving banking connectivity issues and handling online payment settlements under unstable telecommunications infrastructure.
- **GDS & Bank Integrations:** Built direct integrations with domestic acquiring banks, 200+ enterprise merchants, and the Amadeus Global Distribution System (GDS) for atomic flight seat reservations.
- **Master-Slave Replication & CLI Daemons:** Clustered MySQL separating master ACID writes from slave read-reporting, and engineered 1-second CLI background daemon loops to process transaction batches asynchronously.

SOFTWARE ARCHITECT & FULL-STACK DEVELOPER

Independent Technical Consultancy & Retainers

2006 - Present

- **North South Consultants Exchange - NSCE (2025-2026):** Modernized an institutional donor transparency platform for UN, GIZ, and EU development programs. Unified corrupted translation tables into a normalized WordPress Advanced Custom Fields (ACF) schema, built dual-tier spatial governorate filtering for Egypt, and optimized MySQL queries to increase organic search visibility to over 20,000 monthly impressions.
 - **Greek Community of Alexandria - EKA Modernization (2026):** Modernized a 15-year monolithic WordPress portal into a lightweight Full Site Editing (FSE) block theme. Built an AST shortcode migration pipeline, automated 15 years of legacy newsletter date parsing, and configured Nginx reverse proxy media streaming.
 - **barincairo.com (2026):** Architected a decoupled Web GIS platform utilizing React 19, Python 3.12 FastAPI, and PostgreSQL/PostGIS. Built sub-second viewport vector tile streaming, spatial route calculations, and an automated 3-tier venue ingestion pipeline with proximity deduplication.
 - **Saaed Automotive Insurance via Softech (2018):** Stabilized an on-premise CentOS WordPress Multisite insurance portal. Backported and pinned legacy RPM packages to eliminate host crashes, and refactored multi-tenant RBAC permissions to isolate data across agency nodes.
 - **MITCHDesigns (2009-2014):** Embedded Software Architect for a digital agency. Created the core PHP MVC framework and dynamic contract pricing engines (Smash Holidays, Agazat Masr), designed MySQL indexed database views, and trained development teams in Git workflows.
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EDUCATION & FOUNDATIONS

- **B.Sc. in Business Information Systems (BIS)** | Arab Academy for Science, Technology & Maritime Transport (AASTMT)
- **Completed 5-Year Engineering Curriculum Coursework:** Operating Systems Design, Relational Database Theory, Algorithms & Data Structures, SCADA Systems.
- **Interactive Engineering Vault & Case Studies:** Complete 20-year multi-project archive and detailed architectural use cases available at alexseif.com/case-studies

Single Source of Truth: Complete 20-year multi-project architectural vault, systems diagrams, and engineering dossiers available at: [**https://alexseif.com/case-studies**](https://alexseif.com/case-studies)